ExamGOAL
PYQ PDF

Free · With answer key

NEET · PHYSICS · CHAPTER-WISE PYQS

Moving Charges and Magnetism

NEET Physics · Previous Year Questions with Answer Key

80	36	2000 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

MCQ (Single Correct Answer)	80 questions
Answer key	at the end

Practise online

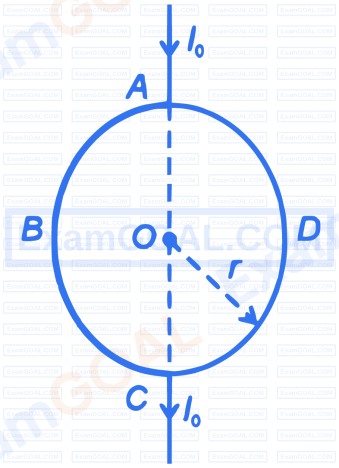
Every question with its full solution, free on **questions.examside.com**

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

MCQ (Single Correct Answer) 80 questions

Q1. RE-NEET 2026

A current l_0 flows through a metallic circular loop of radius r as shown in the figure. Resistance of the segment ABC is half that of ADC. Magnitude of magnetic field at the centre O of the loop is :



Open solution

(A) $\frac{\mu_0 I_0}{2\pi r}$ (B) $\frac{\mu_0 I_0}{12r}$ (C) $\frac{\mu_0 I_0}{4r}$ (D) $\frac{\mu_0 I_0}{2r}$

Q2. RE-NEET 2026

Two infinitely long parallel conducting wires A and B carry currents I and $2I$, respectively, in the same direction. The wire A has uniform mass per unit length λ and lies on an insulated floor. The wire B is kept fixed at a height h above the floor. The minimum magnitude of h so that the wire A does not rise from the floor is: [g is the acceleration due to gravity and μ_0 is the permeability of free space.]

Open solution

(A) $\frac{4\mu_0 I^2}{\pi\lambda g}$ (B) $\frac{\mu_0 I^2}{2\pi\lambda g}$
(C) $\frac{\mu_0 I^2}{\pi\lambda g}$ (D) $\frac{2\mu_0 I^2}{\pi\lambda g}$

Q3. NEET 2026

A 100-turn closely wound circular coil of radius 5 cm has a magnetic field of 3.14×10^{-3} T at its centre. The current flowing through the coil, and the magnitude of the magnetic moment of this coil are, respectively :

(Take $\mu_0 = 4\pi \times 10^{-7}$ T m/A)

Open solution

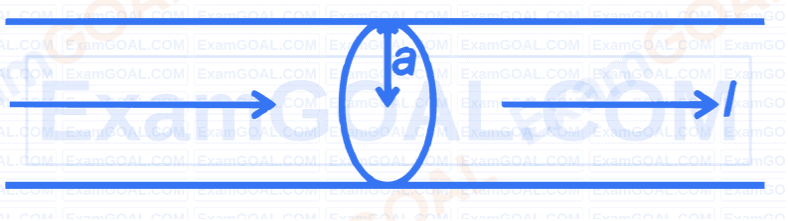
(A) 2 A, 10 A m^2 (B) 2.5 A, 20 A m^2
(C) 2 A, 4 A m^2 (D) 2.5 A, 2 A m^2

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

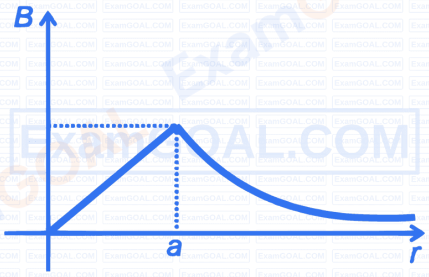
Q4. NEET 2026

Open solution

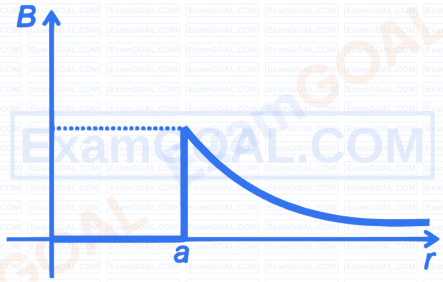
The figure given below shows a long straight solid wire of circular cross-section of radius ' a ' carrying steady current I. The current I is uniformly distributed across its cross-section. The plot which correctly represents the variation of magnetic field (B) with distance (r) from the axis of the conductor in the region is :



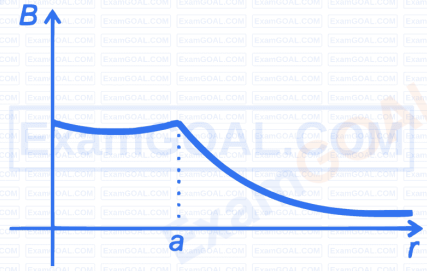
(A)



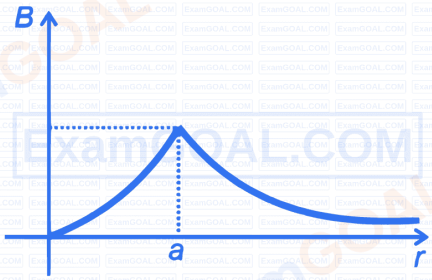
(B)



(C)



(D)



Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q5. NEET 2025

A parallel plate capacitor made of circular plates is being charged such that the surface charge density on its plates is increasing at a constant rate with time. The magnetic field arising due to displacement current is:

(A) Non-zero everywhere with maximum at the imaginary cylindrical surface connecting peripheries of the plates

(B) Zero between the plates and non-zero outside

(C) Zero at all places

(D) Constant between the plates and zero outside the plates

Open solution

Q6. NEET 2025

A model for quantized motion of an electron in a uniform magnetic field B states that the flux passing through the orbit of the electron is $n(h/e)$ where n is an integer, h is Planck's constant and e is the magnitude of electron's charge. According to the model, the magnetic moment of an electron in its lowest energy state will be (m is the mass of the electron)

(A) $\frac{heB}{\pi m}$ (B) $\frac{heB}{2\pi m}$ (C) $\frac{he}{\pi m}$ (D) $\frac{he}{2\pi m}$

Open solution

Q7. NEET 2025

An electron (mass 9×10^{-31} kg and charge 1.6×10^{-19}C) moving with speed $c/100 (c =$ speed of light) is injected into a magnetic field $\vec{B}$ of magnitude 9×10^{-4} T perpendicular to its direction of motion. We wish to apply a uniform electric field $\vec{E}$ together with the magnetic field so that the electron does not deflect from its path. Then (speed of light $c = 3 \times 10^8$ ms^{-1})

(A) $\vec{E}$ is parallel to $\vec{B}$ and its magnitude is 27×10^2 V m^{-1}

(B) $\vec{E}$ is parallel to $\vec{B}$ and its magnitude is 27×10^4 V m^{-1}

(C) $\vec{E}$ is perpendicular to $\vec{B}$ and its magnitude is 27×10^4 V m^{-1}

(D) $\vec{E}$ is perpendicular to $\vec{B}$ and its magnitude is 27×10^2 V m^{-1}

Open solution

Q8. NEET 2025

A 2 amp current is flowing through two different small circular copper coils having radii ratio $1 : 2$. The ratio of their respective magnetic moments will be

(A) $2 : 1$ (B) $4 : 1$ (C) $1 : 4$ (D) $1 : 2$

Open solution

Q9. NEET 2024

A tightly wound 100 turns coil of radius 10 cm carries a current of 7 A. The magnitude of the magnetic field at the centre of the coil is (Take permeability of free space as $4\pi \times 10^{-7}$SI units):

(A) 44 mT (B) 4.4 T (C) 4.4 mT (D) 44 T

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q10. NEET 2024

A parallel plate capacitor is charged by connecting it to a battery through a resistor. If I is the current in the circuit, then in the gap between the plates:

(A) There is no current

(B) Displacement current of magnitude equal to I flows in the same direction as I

(C) Displacement current of magnitude equal to I flows in a direction opposite to that of I

(D) Displacement current of magnitude greater than I flows but can be in any direction

Open solution

Q11. NEET 2023 Manipur

A long straight wire of length $2\ \mathrm{m}$ and mass $250\ \mathrm{g}$ is suspended horizontally in a uniform horizontal magnetic field of $0.7\ \mathrm{T}$. The amount of current flowing through the wire will be $\left(\mathrm{g} = 9.8\ \mathrm{ms}^{-2}\right)$:

(A) 2.45 A (B) 2.25 A (C) 2.75 A (D) 1.75 A

Open solution

Q12. NEET 2023 Manipur

A uniform electric field and a uniform magnetic field are acting along the same direction in a certain region. If an electron is projected in the region such that its velocity is pointed along the direction of fields, then the electron:

(A) will turn towards right of direction of motion

(B) will turn towards left of direction of motion

(C) speed will decrease

(D) speed will increase

Open solution

Q13. NEET 2023

A wire carrying a current I along the positive x-axis has length L. It is kept in a magnetic field $\overrightarrow{\mathrm{B}} = (2\hat{\mathrm{i}} + 3\hat{\mathrm{j}} - 4\hat{\mathrm{k}})\mathrm{T}$. The magnitude of the magnetic force acting on the wire is :

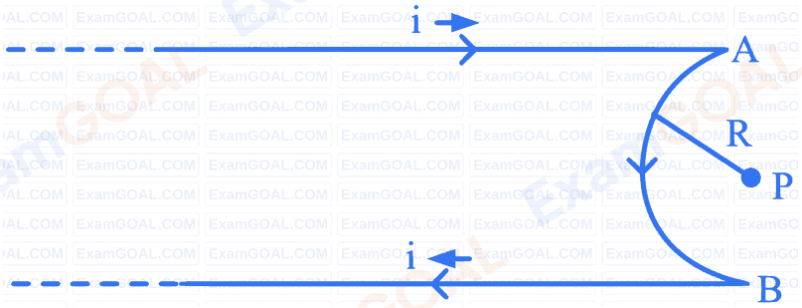
(A) $\sqrt{5}\ \mathrm{IL}$ (B) $5\ \mathrm{IL}$ (C) $\sqrt{3}\ \mathrm{IL}$ (D) 3 IL

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q14. NEET 2023

A very long conducting wire is bent in a semi-circular shape from A to B as shown in figure. The magnetic field at point P for steady current configuration is given by :



(A) $\frac{\mu_0 i}{4R}$ pointed away from the page

(B) $\frac{\mu_0 i}{4R}\left[1-\frac{2}{\pi}\right]$ pointed away from page

(C) $\frac{\mu_0 i}{4R}\left[1-\frac{2}{\pi}\right]$ pointed into the page

(D) $\frac{\mu_0 i}{4R}$ pointed into the page

Open solution

Q15. NEET 2022 Phase 2

A closely packed coil having 1000 turns has an average radius of 62.8 cm. If current carried by the wire of the coil is 1 A, the value of magnetic field produced at the centre of the coil will be (permeability of free space $= 4\pi \times 10^{-7}$ H/m) nearly

(A) 10^{-3} T (B) 10^{-1} T (C) 10^{-2} T (D) 10^{2} T

Open solution

Q16. NEET 2022 Phase 2

The shape of the magnetic field lines due to an infinite long, straight current carrying conductor is

(A) a plane (B) a straight line (C) circular (D) elliptical

Open solution

Q17. NEET 2022 Phase 2

Two very long, straight, parallel conductors A and B carry current of 5 A and 10 A respectively and are at a distance of 10 cm from each other. The direction of current in two conductors is same. The force acting per unit length between two conductors is : ($\mu_0 = 4\pi \times 10^{-7}$ SI unit)

(A) 1×10^{-4} Nm^{-1} and is repulsive (B) 2×10^{-4} Nm^{-1} and is attractive

(C) 2×10^{-4} Nm^{-1} and is repulsive (D) 1×10^{-4} Nm^{-1} and is attractive

Open solution

Q18. NEET 2022 Phase 2

The magnetic field on the axis of a circular loop of radius 100 cm carrying current $I = \sqrt{2}\,A$, at point 1 m away from the centre of the loop is given by :

(A) 6.28×10^{-4} T (B) 3.14×10^{-7} T (C) 6.28×10^{-7} T (D) 3.14×10^{-4} T

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q19. NEET 2022 Phase 1

A long solenoid of radius 1 mm has 100 turns per mm. If 1 A current flows in the solenoid, the magnetic field strength at the centre of the solenoid is

(A) 6.28×10^{-2} T (B) 12.56×10^{-2} T (C) 12.56×10^{-4} T (D) 6.28×10^{-4} T

Open solution

Q20. NEET 2022 Phase 1

Given below are two statements:

Statement I : Biot-Savart's law gives us the expression for the magnetic field strength of an infinitesimal current element (Idl) of a current carrying conductor only.

Statement II : Biot-Savart's law is analogous to Coulomb's inverse square law of charge q, with the former being related to the field produced by a scalar source, Idl while the latter being produced by a vector source, q.

In light of above statements choose the most appropriate answer from the options given below.

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct and Statement II is incorrect

(D) Statement I is incorrect and Statement II is correct

Open solution

Q21. NEET 2022 Phase 1

From Ampere's circuital law for a long straight wire of circular cross-section carrying a steady current, the variation of magnetic field in the inside and outside region of the wire is

(A) Uniform and remains constant for both the regions.

(B) A linearly increasing function of distance upto the boundary of the wire and then linearly decreasing for the outside region.

(C) A linearly increasing function of distance r upto the boundary of the wire and then decreasing one with $\frac{1}{r}$ dependence for the outside region.

(D) A linearly decreasing function of distance upto the boundary of the wire and then a linearly increasing one of the outside region.

Open solution

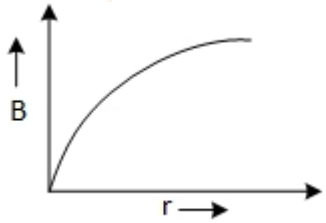
Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q22. NEET 2021

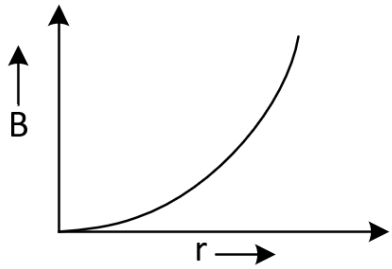
Open solution

A thick current carrying cable of radius 'R' carries current 'I' uniformly distributed across its cross section. The variation of magnetic field B(r) due to the cable with the distance 'r' from the axis of the cable is represented by :

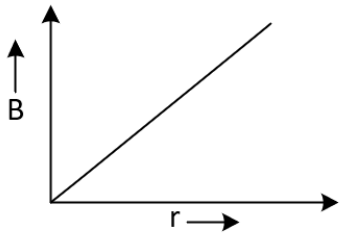
(A)



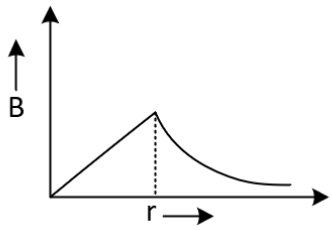
(B)



(C)



(D)

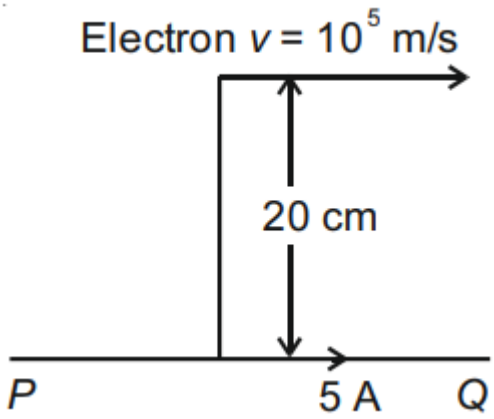


Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q23. NEET 2021

An infinitely long straight conductor carries a current of 5A as shown. An electron is moving with a speed of 10^5 m/s parallel to the conductor. The perpendicular distance between the electron and the conductor is 20cm at an instant. Calculate the magnitude of the force experienced by the electron at that instant.

Open solution



(A) 8×10^{-20} N (B) 4×10^{-20} N (C) $8\pi \times 10^{-20}$ N (D) $4\pi \times 10^{-20}$ N

Q24. NEET 2020 Phase 1

A long solenoid of 50 cm length having 100 turns carries a current of 2.5 A. The magnetic field at the centre of the solenoid is :
$(\mu_0 = 4\pi \times 10^{-7} T m A^{-1})$

Open solution

(A) 3.14 $\times 10^{-4} T$ (B) 6.28 $\times 10^{-5} T$
(C) 3.14 $\times 10^{-5} T$ (D) 6.28 $\times 10^{-4} T$

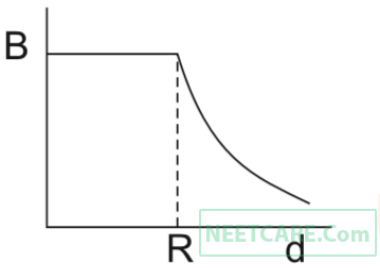
Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q25. NEET 2019

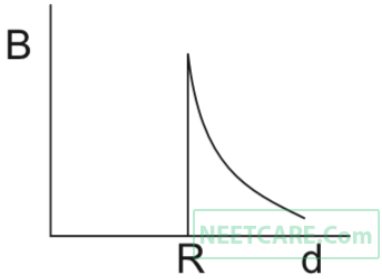
A cylindrical conductor of radius R is carrying a constant current. The plot of the magnitude of the magnetic field B with the distance d from the centre of the conductor, is correctly represented by the figure :

Open solution

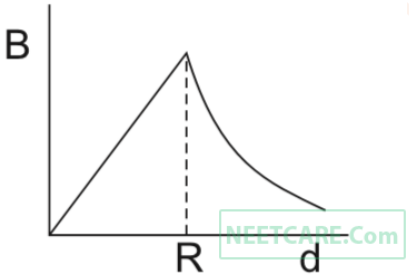
(A)



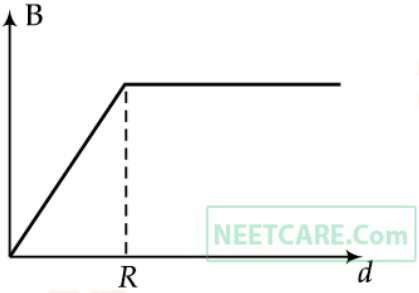
(B)



(C)



(D)



Q26. NEET 2019

Ionized hydrogen atoms and α-particles with same momenta enters perpendicular to a constant magnetic field, B. The ratio of their radii of their paths $r_H : r_\alpha$ will be :

Open solution

(A) 1 : 2 (B) 4 : 1 (C) 1 : 4 (D) 2 : 1

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q27. NEET 2018

A metallic rod of mass per unit length 0.5 kg m^{-1} is lying horizontally on a smooth inclined plane which makes an angle of 30° with the horizontal. The rod is not allowed to slide down by flowing a current through it when a magnetic field of induction 0.25 T is acting on it in the vertical direction. The current flowing in the rod to keep it stationary is

Open solution

(A) 7.14 A (B) 5.98 A (C) 14.76 A (D) 11.32 A

Q28. NEET 2018

Current sensitivity of a moving coil galvanometer is 5 div/mA and its voltage sensitivity (angular deflection per unit voltage applied) is 20 div/V. The resistance of the galvanometer is

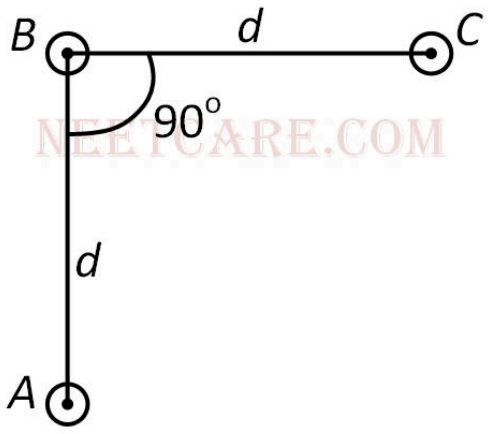
Open solution

(A) 40 Ω (B) 25 Ω (C) 250 Ω (D) 500 Ω

Q29. NEET 2017

An arrangement of three parallel straight wires placed perpendicular to plane of paper carrying same current $'I'$ along the same direction as shown in figure. Magnitude of force per unit length on the middle wire $'B'$ is given by

Open solution



(A) $\frac{2\mu_0 I^2}{\pi d}$ (B) $\frac{\sqrt{2}\mu_0 I^2}{\pi d}$

(C) $\frac{\mu_0 I^2}{\sqrt{2}\pi d}$ (D) $\frac{\mu_0 I^2}{2\pi d}$

Q30. NEET 2016 Phase 2

A long wire carrying a steady current is bent into a circular loop of one turn. The magnetic field at the centre of the loop is B. It is then bent into a circular coil of n turns. The magnetic field at the centre of this coil of n turns will be

Open solution

(A) nB (B) n^2B (C) 2nB (D) $2n^2B$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q31. NEET 2016 Phase 2

An electron is moving in a circular path under the influence of a transverse magnetic field of 3.57×10^{-2} T. If the value of e/m is 1.76×10^{11} C kg^{-1}, the frequency of revolution of the electron is

(A) 1 GHz (B) 100 MHz (C) 62.8 MHz (D) 6.28 MHz

Open solution

Q32. NEET 2016 Phase 1

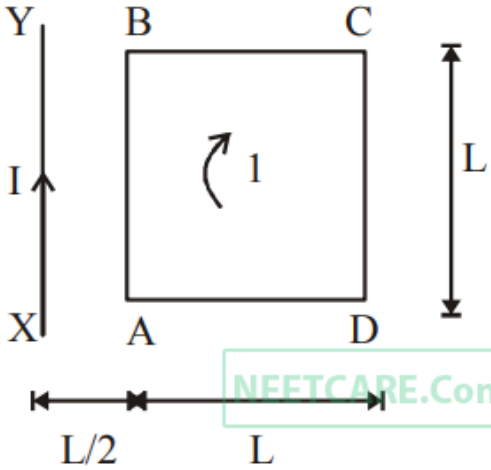
A long straight wire of radius a carries a steady current I. The current is uniformly distributed over its cross-section. The ratio of the magnetic fields B and B', at radial distance $\frac{a}{2}$ and $2a$ respectively, from the axis of the wire is

(A) 1 (B) 4 (C) $\frac{1}{4}$ (D) $\frac{1}{2}$

Open solution

Q33. NEET 2016 Phase 1

A square loop ABCD carrying a current i, is placed near and coplanar with a long straight conductor XY carrying a current I, the net force on the loop will be



(A) $\frac{2\mu_0 IiL}{3\pi}$ (B) $\frac{\mu_0 IiL}{2\pi}$ (C) $\frac{2\mu_0 Ii}{3\pi}$ (D) $\frac{\mu_0 Ii}{2\pi}$

Open solution

Q34. AIPMT 2015

A proton and an alpha particle both enter a region of uniform magnetic field B, moving at right angles to the field B. If the radius of circular orbits for both the particles is equal and the kinetic energy acquired by proton is 1 MeV, the energy acquired by the alpha particle will be

(A) 1.5 MeV (B) 1 MeV (C) 4 MeV (D) 0.5 MeV

Open solution

Q35. AIPMT 2015 Cancelled Paper

An electron moving in a circular orbit of radius r makes n rotations per second. The magnetic field produced at the centre has magnitude

(A) $\frac{\mu_0 n^2 e}{r}$ (B) $\frac{\mu_0 ne}{2r}$ (C) $\frac{\mu_0 ne}{2\pi r}$ (D) Zero

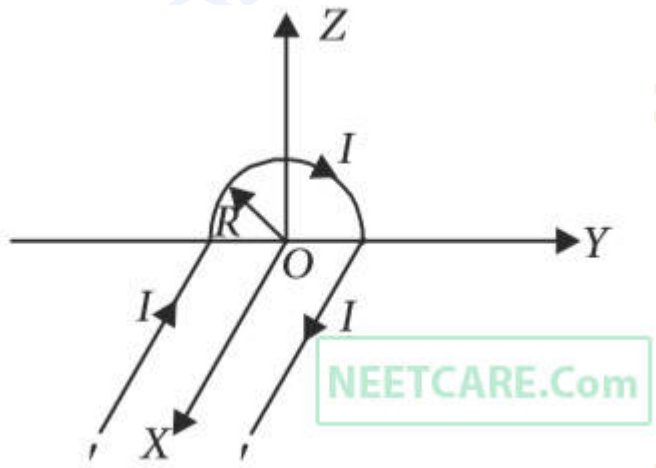
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q36. AIPMT 2015 Cancelled Paper

Open solution

A wire carrying current I has the shape shown in adjoining figure.



NEETCARE.Com

Linear parts of the wire are very long and parallel to X-axis while semicircular protion of radius R is lying in Y-Z plane. Magtnetic field at pont O is

(A) $\vec{B} = -\frac{\mu_0 I}{4\pi R}\left(\pi\hat{i} + 2\hat{k}\right)$

(B) $\vec{B} = \frac{\mu_0 I}{4\pi R}\left(\pi\hat{i} - 2\hat{k}\right)$

(C) $\vec{B} = \frac{\mu_0 I}{4\pi R}\left(\pi\hat{i} + 2\hat{k}\right)$

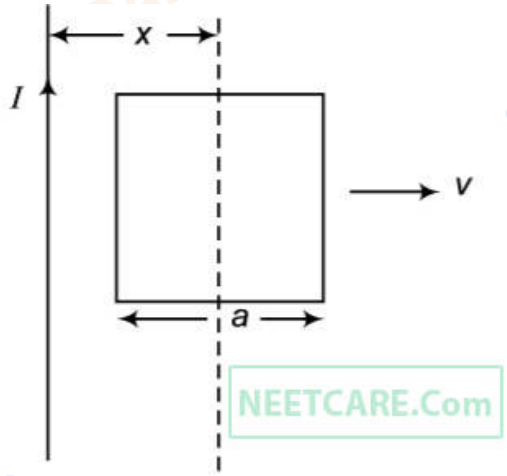
(D) $\vec{B} = -\frac{\mu_0 I}{4\pi R}\left(\pi\hat{i} - 2\hat{k}\right)$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q37. AIPMT 2015 Cancelled Paper

A conducting square frame of side 'a' and a long straight wire carrying current I are located in the same plane as shown in the figure. The frame moves to the right with a constant velocity 'V'. The emf induced in the frame will be proportional to

Open solution



(A) $\frac{1}{(2x+a)^2}$

(B) $\frac{1}{(2x-a)(2x+a)}$

(C) $\frac{1}{x^2}$

(D) $\frac{1}{(2x-a)^2}$

Q38. AIPMT 2014

In an ammeter 0.2% of main current passes through the galvanometer. If resistance of galvanometer is G, the resistance of ammeter will be

Open solution

(A) $\frac{1}{499}G$

(B) $\frac{499}{500}G$

(C) $\frac{1}{500}G$

(D) $\frac{500}{499}G$

Q39. AIPMT 2014

Two identical long conducting wires AOB and COD are placed at right angle to each other, with one above other such that O is their common point for the two. The wires carry I_1 and I_2 currents, respectively. Point P is lying at distance f from O along a direction perpendicular to the plane containing the wires. The magnetic field at the point P will be

Open solution

(A) $\frac{\mu_0}{2\pi d}\left(\frac{I_1}{I_2}\right)$

(B) $\frac{\mu_0}{2\pi d}(I_1+I_2)$

(C) $\frac{\mu_0}{2\pi d}\left(I_1^2-I_2^2\right)$

(D) $\frac{\mu_0}{2\pi d}\left(I_1^2+I_2^2\right)^{1/2}$

Q40. NEET 2013 (Karnataka)

A long straight wire carries a certain current and produces a magnetic field 2×10^{-4} Wb m^{-2} at a perpendicular distance of 5 cm from the wire. An electron situated at 5 cm from the wire moves with a velocity 10^7 m/s towards the wire along perpendicular to it. The force experienced by the electron will be (charge on electron 1.6×10^{-19} C)

Open solution

(A) 3.2 N

(B) 3.2×10^{-16} N

(C) 1.6×10^{-16} N

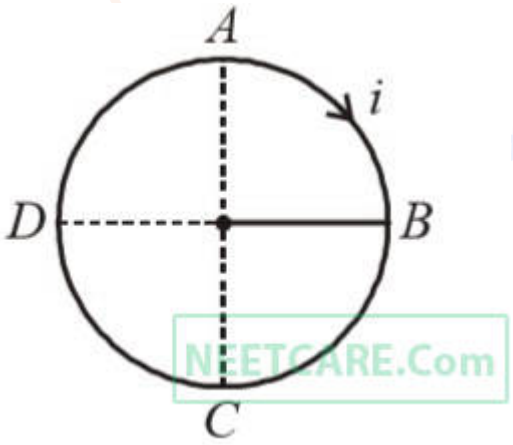
(D) zero

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q41. NEET 2013 (Karnataka)

A circular coil ABCD carrying a current 'i' is placed in a uniform magnetic field. If the magnetic force on the segment AB is $\vec{F}$, the force on the remaining segment BCDA is

Open solution



(A) $-\vec{F}$
(B) $3\vec{F}$
(C) $-3\vec{F}$
(D) $\vec{F}$

Q42. NEET 2013

When a proton is released from rest in a room, it starts with an initial acceleration a_0 towards west. When it is projected towards north with a speed v_0 it moves with an initial acceleration $3a_0$ towards west. The electric and magnetic fields in the room are

Open solution

(A) $\frac{ma_0}{e}$ east, $\frac{3ma_0}{ev_0}$ up
(B) $\frac{ma_0}{e}$ east, $\frac{3ma_0}{ev_0}$ down
(C) $\frac{ma_0}{e}$ west, $\frac{2ma_0}{ev_0}$ up
(D) $\frac{ma_0}{e}$ west, $\frac{2ma_0}{ev_0}$ down

Q43. AIPMT 2012 Mains

A proton carrying 1 MeV kinetic energy is moving in a circular path of radius R in uniform magnetic field. What should be the energy of an α-particle to describe a circle of same radius in the same field?

Open solution

(A) 2 MeV
(B) 1 MeV
(C) 0.5 MeV
(D) 4 MeV

Q44. AIPMT 2012 Prelims

Two similar coils of radius R are lying concentrically with their planes at right angles to each other. The currents flowing in them are I and $2I$, respectively. The resultant magnetic field induction at the centre will be

Open solution

(A) $\frac{\sqrt{5}\mu_0 I}{2R}$
(B) $\frac{\sqrt{5}\mu_0 I}{R}$
(C) $\frac{\mu_0 I}{2R}$
(D) $\frac{\mu_0 I}{R}$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q45. AIPMT 2012 Prelims

A milli voltmeter of 25 milli volt range is to be converted into an ammeter of 25 ampare range. The value (in ohm) of neccessary shunt will be

(A) 0.001 (B) 0.01 (C) 1 (D) 0.05

Open solution

Q46. AIPMT 2012 Prelims

An alternating electric field, of frequency v, is applied across the dees (radius = R) of a cyclotron that is being used to accelerate protons (mass = m). The operating magnetic field (B) used in the cyclotron and the kinetic energy (K) of the proton beam, produced by it, are given by

(A) $B = \frac{mv}{e}$ and $K = 2m\pi^2v^2R^2$ (B) $B = \frac{2\pi mv}{e}$ $K = m^2\pi vR^2$

(C) $B = \frac{2\pi mv}{e}$ $K = 2m\pi^2v^2R^2$ (D) $B = \frac{mv}{e}$ $K = m^2\pi vR^2$

Open solution

Q47. AIPMT 2011 Mains

A galvanometer of resistance, G, is shunted by a resistance S ohm. To keep the main current in the circuit unchanged, the resistance to be put in series with the galvanometer is

(A) $\frac{G}{(S+G)}$ (B) $\frac{S^2}{(S+G)}$

(C) $\left(\frac{SG}{S+G}\right)$ (D) $\frac{G^2}{(S+G)}$

Open solution

Q48. AIPMT 2011 Mains

Charge q is uniformly spread on a thin ring of radius R. The ring rotates about its axis with a uniform frequency f Hz. The magnitude of magnetic induction at the center of the ring is

(A) $\frac{\mu_0 qf}{2\pi R}$ (B) $\frac{\mu_0 qf}{2R}$ (C) $\frac{\mu_0 q}{2fR}$ (D) $\frac{\mu_0 q}{2\pi fR}$

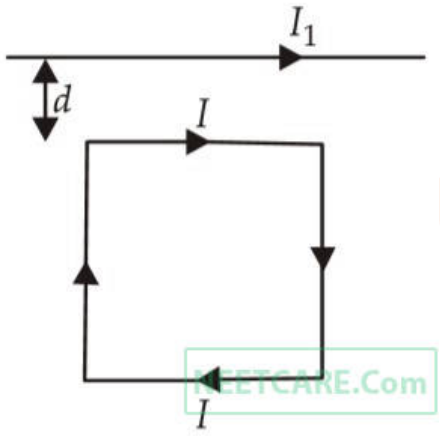
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q49. AIPMT 2011 Mains

A square loop, carrying a steady current I, is placed in a horizontal plane near a long straight conductor carrying a steady current I_1 at a distance d from the conductor as shown in figure. The loop will experience

Open solution



(A) a net attractive force towards the conductor

(B) a net repulsive force away from the conductor

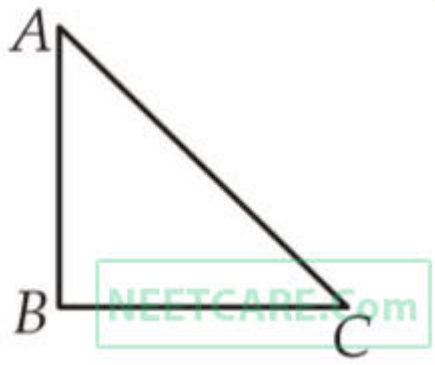
(C) a net torque acting upward perpendicular to the horizontal plane

(D) a net torque acting downward normal to horizontal plane

Q50. AIPMT 2011 Prelims

A current carrying closed loop in the form of a right angle isosceles triangle ABC is placed in uniform magnetic field acting along AB. If the magnetic force on the arm BC is $\overrightarrow{F}$, the force on the arm AC is

Open solution



(A) $-\sqrt{2}\overrightarrow{F}$ (B) $-\overrightarrow{F}$ (C) $\overrightarrow{F}$ (D) $\sqrt{2}\overrightarrow{F}$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q51. AIPMT 2011 Prelims

A uniform electric field and a uniform magnetic field are acting along the same direction in a certain region. If an electron is projected in the region such that its velocity is pointed along the direction of fields, then the electron

Open solution

(A) will turn towards right of direction of motion

(B) speed will decrease

(C) speed will increase

(D) will turn towards left of direction of motion

Q52. AIPMT 2010 Mains

A current loop consists of two identical semicircular parts each of radius R, one lying in the x-y plane and the other in x-z plane. If the current in the loop is i. The resultant magnetic field due to the two semicircular parts at their common centre is

Open solution

(A) $\frac{\mu_0 i}{2\sqrt{2}R}$ (B) $\frac{\mu_0 i}{2R}$ (C) $\frac{\mu_0 i}{4R}$ (D) $\frac{\mu_0 i}{\sqrt{2}R}$

Q53. AIPMT 2010 Mains

A closely wound solenoid of 2000 turns and area of cross-section 1.5×10^{-4} m^2 carries a current of 2.0 A. It is suspended through its centre and perpendicular to its length, allowing it to turn in a horizontal plane in a uniform magnetic field 5×10^{-2} tesla making an angle of 30° with the axis of the solenoid. The torque on the solenoid will be

Open solution

(A) 3×10^{-3} N m

(B) 1.5×10^{-3} N m

(C) 1.5×10^{-2} N m

(D) 3×10^{-2} N m

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q54. AIPMT 2010 Mains

A particle having a mass of 10^{-2} kg carries a charge of 5×10^{-8} C. The particle is given an initial horizontal velocity of 10^5 m s^{-1} in the presence of electric field $\vec{E}$ and magnetic field $\vec{B}$. To keep the particle moving in a horizontal direction, it is necessary that

(1) $\vec{B}$ should be perpendicular to the direction of velocity and $\vec{E}$ should be along the direction of velocity

(2) Both $\vec{B}$ and $\vec{E}$ should be along the direction of velocity

(3) Both $\vec{B}$ and $\vec{E}$ are mutually perpendicular and perpendicular to the direction of velocity.

(4) $\vec{B}$ should be along the direction of velocity and $\vec{E}$ should be perpendicular to the direction of velocity

Which one of the following pairs of statements is possible ?

(A) (1) and (3) (B) (3) and (4) (C) (2) and (3) (D) (2) and (4)

Open solution

Q55. AIPMT 2010 Prelims

Charge q is uniformly spread on a thin ring of radius R. The ring rotates about its axis with a uniform frequency f Hz. The magnitude of magnetic induction at the center of the ring is

(A) $\frac{\mu_0 qf}{2\pi R}$ (B) $\frac{\mu_0 qf}{2R}$ (C) $\frac{\mu_0 q}{2fR}$ (D) $\frac{\mu_0 q}{2\pi fR}$

Open solution

Q56. AIPMT 2010 Prelims

A galvanometer has a coil of resistance 100 ohm and gives a full scale deflection for 30 mA current. If it is to work as a voltmeter of 30 volt range, the resistance required to be added will be

(A) 900 Ω (B) 1800 Ω (C) 500 Ω (D) 1000 Ω

Open solution

Q57. AIPMT 2010 Prelims

A galvanometer has a coil of resistance 100 ohm and gives a full scale deflection for 30 mA current. If it is to work as a voltmeter of 30 volt range, the resistance required to be added will be

(A) 900 Ω (B) 1800 Ω (C) 500 Ω (D) 1000 Ω

Open solution

Q58. AIPMT 2010 Prelims

A square current carrying loop is suspended in a uniform magnetic field acting in the plane of the loop. If the force on one arm of the loop is $\vec{F}$, the net force on the remaining three arms of the loop is

(A) $3\vec{F}$ (B) $-\vec{F}$ (C) $-3\vec{F}$ (D) $\vec{F}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q59. AIPMT 2009

A galvanometer havings a coil resistance of 60 Ω shows full scale deflection when a current of 1.0 amp passes through it. It can be converted into an ammeter to read currents upto 5.0 amp by

(A) putting in series a resistance of 15 Ω

(B) putting in series a resistance of 240 Ω

(C) putting in parallel a resistance of 15 Ω

(D) putting in parallel a resistance of 240 Ω

Open solution

Q60. AIPMT 2009

The magnetic force acting on a charged particle of charge -2 μC in a magnetic field of 2 T acting in y direction, when the particle velocity is $\left(2\hat{i}+3\hat{j}\right)\times 10^{6}\, ms^{-1}$, is

(A) 4 N in z direction

(B) 8 N in y direction

(C) 8 N in z direction

(D) 8 N in $-$ z direction

Open solution

Q61. AIPMT 2009

Under the influence of a uniform magnetic field, a charged particle moves with constant speed v in a circle of radius R. The time period of rotation of the particle

(A) depends on R and not on v

(B) is independent of both v and R

(C) depends on both v and R

(D) depends on v and not on R

Open solution

Q62. AIPMT 2008

A particle of mass m, charge Q and kinetic energy T enters a transverse uniform magnetic field of induction $\overrightarrow{B}$. After 3 seconds the kinetic energy of the particle will be

(A) T

(B) 4 T

(C) 3 T

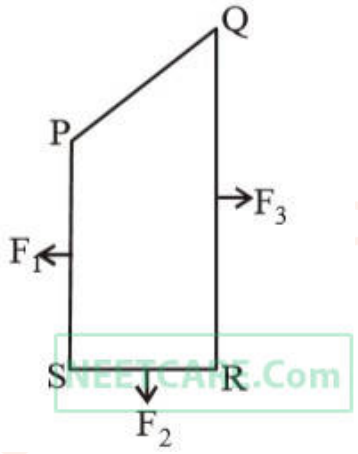
(D) 2 T

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q63. AIPMT 2008

A closed loop PQRS carrying a current is placed in a uniform magnetic field. If the magnetic forces on segments PS, SR and RQ are F_1, F_2 and F_3 respectively and are in the plane of the paper and along the directions shown, the force on the segment QP is



Open solution

(A) $\sqrt{(F_3 - F_1)^2 - F_2^2}$

(B) $F_3 - F_1 + F_2$

(C) $F_3 - F_1 - F_2$

(D) $\sqrt{(F_3 - F_1)^2 + F_2^2}$

Q64. AIPMT 2008

A galvanometer of resistance 50 Ω is connected to a battery of 3 V along with a resistance of 2950 Ω in series. A full scale deflection of 30 divisions is obtained in the galvanometer. In order to reduce this deflection to 20 divisions, the resistance in series should be

Open solution

(A) 6050 Ω

(B) 4450 Ω

(C) 5050 Ω

(D) 5550 Ω

Q65. AIPMT 2007

Under the influence of a uniform magnetic field, a charged particle moves with constant speed v in a circle of radius R. The time period of rotation of the particle

Open solution

(A) depends on R and not on v

(B) is independent of both v and R

(C) depends on both v and R

(D) depends on v and not on R

Q66. AIPMT 2007

The resistance of an ammeter is 13 Ω and its scale is graduated for a current upto 100 amps. After an additional shunt has been connected to this ammeter it becomes possible to measure currents upto 750 amperes by this meter. The value of shunt-resistance is

Open solution

(A) 2 Ω

(B) 0.2 Ω

(C) 2 kΩ

(D) 20 Ω

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q67. AIPMT 2006

When a charged particle moving with velocity $\vec{v}$ is subjected to a magnetic field of induction $\vec{B}$, the force on it is non-zero. This implies that

(A) angle between is either zero or 180°

(B) angle between is necessarily 90°

(C) angle between can have any value other than 90°

(D) angle between can have any value other than zero and 180°.

Open solution

Q68. AIPMT 2006

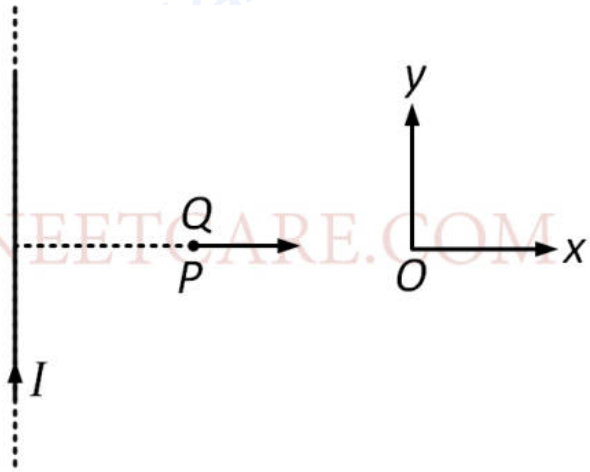
Two circular coils 1 and 2 are made from the same wire but the radius of the 1st coil is twice that of the 2nd coil. What ratio of potential difference in volts should be applied across them so that the magnetic field at their centres is the same?

(A) 2 (B) 3 (C) 4 (D) 6

Open solution

Q69. AIPMT 2005

A very long straight wire carries a current I. At the instant when a charge +Q at point P has velocity $\vec{v}$, as shown, the force on the charge is



(A) along Oy (B) opposite to Oy (C) along Ox (D) opposite to Ox

Open solution

Q70. AIPMT 2005

An electron moves in a circular orbit with a uniform speed v. It produces a magnetic field B at the centre of the circle. The radius of the circle is proportional to

(A) $\sqrt{B/v}$ (B) B/v (C) $\sqrt{v/B}$ (D) v/B

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q71. AIPMT 2004

A galvanometer of 50 ohm resistance has 25 divisions. A current of 4×10^{-4} ampere gives a deflection of one division. To convert this galvanometer into a voltmeter having a range of 25 volts, it should be connected with a resistance of

Open solution

(A) $2500\,\Omega$ as a shunt
(B) $2450\,\Omega$ as a shunt
(C) $2550\,\Omega$ in series
(D) $2450\,\Omega$ in series

Q72. AIPMT 2004

To convert a galvanometer into a voltmeter one should connect a

Open solution

(A) high resistance in series with galvanometer
(B) low resistance in series with galvanometer
(C) high resistance in parallel wilh galvanometer
(D) low resistance in parallel with galvanometer.

Q73. AIPMT 2003

A charged particle moves through a magnetic field in a direction perpendicular to it. Then the

Open solution

(A) speed of the particle remains unchanged
(B) direction of the particle remains unchanged
(C) acceleration remains unchanged
(D) velocity remains unchanged

Q74. AIPMT 2003

A long solenoid carrying a current produces a magnetic field B along its axis. If the current is doubled and the number of turns per cm is halved, the new value of the magnetic field is

Open solution

(A) B/2 (B) B (C) 2B (D) 4B

Q75. AIPMT 2002

To convert a galvanometer into a voltmeter one should connect a

Open solution

(A) high resistance in series with galvanometer
(B) low resistance in series with galvanometer
(C) high resistance in parallel with galvanometer
(D) low resistance in parallel with galvanometer.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q76. AIPMT 2002

A charge q moves in a region where electric field and magnetic field both exist, then force on it is

(A) $q\left(\overrightarrow{v}\times\overrightarrow{B}\right)$

(B) $q\overrightarrow{E}+q\left(\overrightarrow{v}\times\overrightarrow{B}\right)$

(C) $q\overrightarrow{E}+\overrightarrow{q}\left(\overrightarrow{B}\times\overrightarrow{v}\right)$

(D) $q\overrightarrow{B}+\overrightarrow{q}\left(\overrightarrow{E}\times\overrightarrow{v}\right)$

Open solution

Q77. AIPMT 2002

The magnetic field of given length of wire for single turn coil at its centre is B then its value for two turns coil for the same wire is

(A) B/4 (B) B/2 (C) 4B (D) 2B

Open solution

Q78. AIPMT 2001

If number of turns, area and current through a coil is given by n, A and i respectively then its magnetic moment will be

(A) niA (B) n^2iA (C) niA^2 (D) $\frac{ni}{\sqrt{A}}$

Open solution

Q79. AIPMT 2001

An electron having mass m and kinetic energy E enters a uniform magnetic field B perpendicularly, then its frequency will be

(A) $\frac{eE}{qvB}$ (B) $\frac{2\pi m}{eB}$ (C) $\frac{eB}{2\pi m}$ (D) $\frac{2m}{eBE}$

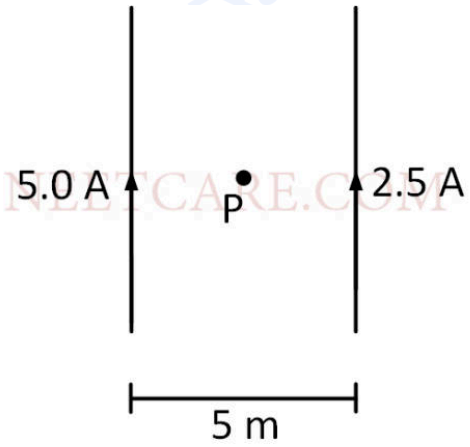
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q80. AIPMT 2000

The magnetic field at centre, P will be

Open solution



(A) $\frac{\mu_0}{4\pi}$ (B) $\frac{\mu_0}{\pi}$ (C) $\frac{\mu_0}{2\pi}$ (D) $4\mu_0\pi$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

1. **B**	2. **C**	3. **D**	4. **A**	5. **A**
6. **D**	7. **D**	8. **C**	9. **C**	10. **B**
11. **D**	12. **C**	13. **B**	14. **B**	15. **A**
16. **C**	17. **D**	18. **B**	19. **B**	20. **C**
21. **C**	22. **D**	23. **A**	24. **D**	25. **C**
26. **D**	27. **D**	28. **C**	29. **C**	30. **B**
31. **A**	32. **A**	33. **C**	34. **B**	35. **B**
36. **A**	37. **B**	38. **C**	39. **D**	40. **B**
41. **A**	42. **D**	43. **B**	44. **A**	45. **A**
46. **C**	47. **D**	48. **B**	49. **A**	50. **B**
51. **B**	52. **A**	53. **C**	54. **C**	55. **B**
56. **A**	57. **A**	58. **B**	59. **C**	60. **D**
61. **B**	62. **A**	63. **D**	64. **B**	65. **B**
66. **A**	67. **D**	68. **C**	69. **A**	70. **C**
71. **D**	72. **A**	73. **A**	74. **B**	75. **A**
76. **B**	77. **C**	78. **A**	79. **C**	80. **C**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs